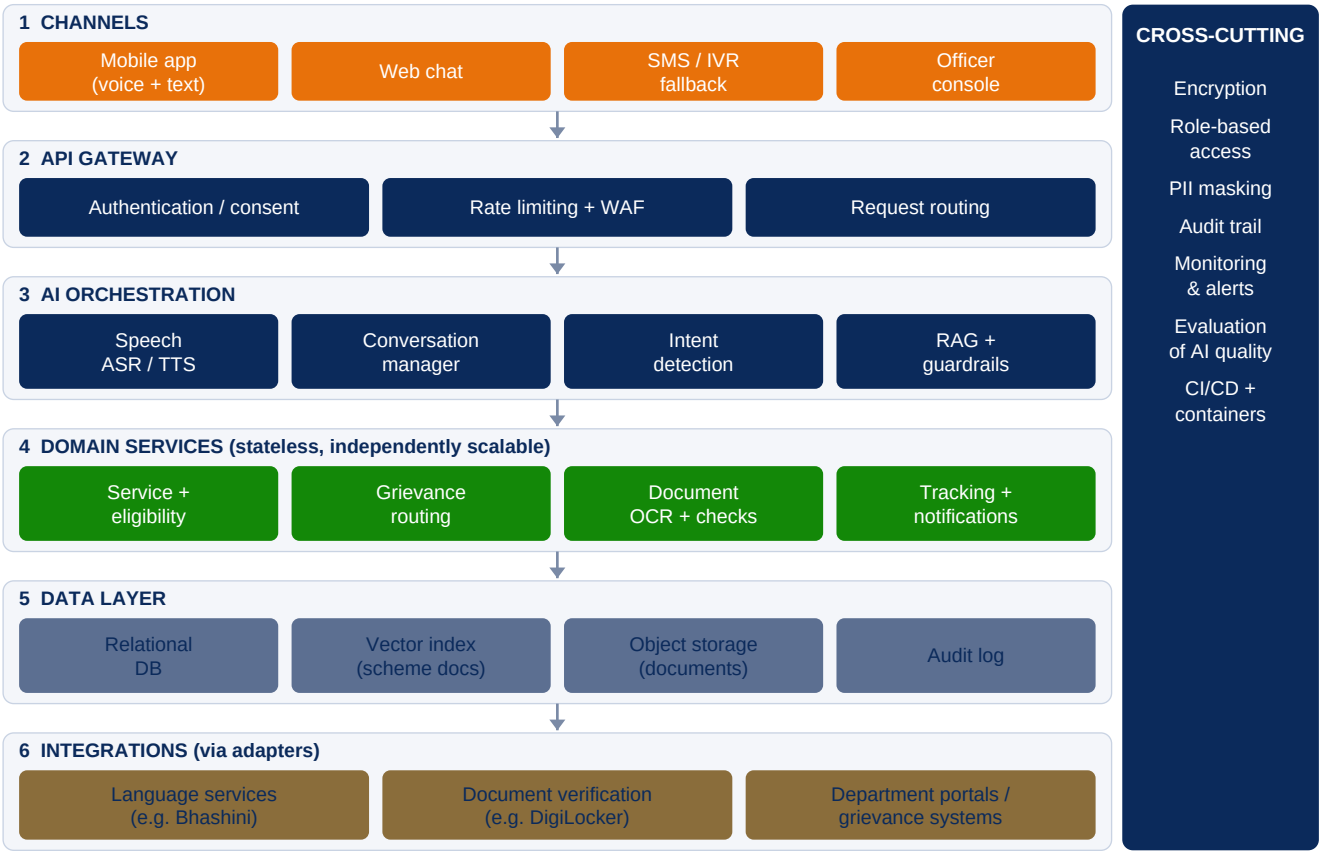
Approach in brief: a layered, **modular and stateless** architecture. A thin AI orchestration layer sits on top of independent domain services (eligibility, grievance routing, OCR, tracking). Models and external systems are reached through **adapters**, so any model, language service or department system can be replaced without redesign. Security and human oversight are built into every layer, not added later.

1. Architecture overview



2. Design principles

Modular	Each capability is a separate service behind an API, so one module can be improved, scaled or replaced independently.
Model-agnostic	The language model, speech and OCR engines sit behind adapters; no vendor lock-in and the best option per language can be chosen.
Grounded AI	Answers come from verified scheme and policy documents (retrieval), not from the model's memory alone, with the source shown.
Human in the loop	Confidence thresholds and sensitive-case rules route to an officer with full context.
Privacy by design	Collect minimum data, mask personal data before it reaches any model, keep an audit trail of every action.

3. Technology choices and rationale

Component	Choice (examples)	Rationale
Mobile / web	Cross-platform framework (e.g. Flutter or React Native); responsive web chat	One codebase for Android and iOS; Android reach matters most for citizens
Backend / APIs	Containerised services in Python (AI-heavy) and a standard web framework; REST APIs	Python fits the AI ecosystem; containers keep services portable across clouds
Language model	Hosted or self-hosted LLM behind an adapter; smaller classifier models for intent	Use a small, cheap model for routine intent and a larger one only when needed
Retrieval (RAG)	Embedding model plus vector index (e.g. pgvector or a dedicated vector store)	Keeps answers tied to official documents and updatable without retraining
Speech	ASR and TTS for Indian languages (e.g. national language services such as Bhashini, or cloud speech APIs)	Enables voice-first access; multiple providers behind one interface
OCR / document AI	Open-source OCR (e.g. Tesseract) or cloud OCR, plus rule-based field validation	Proven; rules come from department checklists so results are explainable
Data stores	PostgreSQL for records; object storage for documents; Redis cache; message queue for async jobs	Mature, well-understood, widely available in government-approved clouds
Notifications	SMS gateway and in-app push	Reaches users without smartphones or data
Hosting / DevOps	Containers with an orchestrator; CI/CD; deployable on a government-empanelled cloud or on-premise as policy requires	Meets data-residency needs; same build runs anywhere

4. How a request flows

1	Citizen speaks or types a need; speech is converted to text in the citizen's language.
2	Gateway authenticates the user and records consent; personal data is masked before AI processing.
3	Intent detection decides: service query, grievance, or document task; the retrieval step finds the relevant official content.
4	The matching domain service runs: eligibility check, grievance routing, or OCR and validation of uploaded documents.
5	The answer is returned with sources and next steps; any action (apply, file, upload) is stored with a tracking ID.
6	If confidence is low or the case is sensitive, it is escalated to an officer with the full conversation; status updates are sent automatically.

5. Scalability

Horizontal scaling	Stateless services run as multiple container replicas behind a load balancer and scale automatically with demand.
Asynchronous work	OCR, notifications and heavy analysis go through a message queue so spikes do not slow the conversation.
Cost-aware AI	Small models and cached answers handle common queries; the larger model is used only for complex ones; frequent questions are answered from the knowledge base.
Data growth	Read replicas, indexed search and archival of closed cases; documents in object storage rather than the database.
Multi-department / multi-state	A new department is onboarded by adding its documents, rules and routing table (configuration), not by changing the core code.
Resilience	Health checks, retries, graceful degradation: if AI is unavailable the app falls back to guided menus, SMS and human helpline.

6. Security, privacy and responsible AI

Area	Controls
Data protection	Encryption in transit (TLS) and at rest; data minimisation; retention limits; documents deletable on request.
Access control	Role-based access for officers and admins; least privilege; multi-factor for staff accounts; session management.
Privacy and consent	Explicit consent before documents are processed; personal identifiers masked before model calls; design aligned with India's Digital Personal Data Protection Act, 2023.
Auditability	Immutable audit log of every access, AI decision and officer action; supports grievance and accountability review.
Application security	Input validation, rate limiting and WAF; secure upload scanning; dependency scanning; vulnerability assessment and penetration testing before pilot.
AI-specific safety	Guardrails against prompt injection and off-topic or harmful output; answers limited to approved sources; confidence threshold with human escalation.
Fairness and quality	Regular evaluation on test question sets per language; track accuracy, routing accuracy and escalation rate; human review of samples.
Hosting and residency	Deployed per government hosting policy so citizen data stays within approved infrastructure.

7. Observability and operations

Central logging, metrics and alerting cover latency, error rates, AI answer quality and queue depth. Staged deployments (test, pilot, production) with automated tests and rollback keep releases safe. Prompts, models and knowledge-base versions are tracked so any answer can be traced and reproduced.

Note: technologies named are representative options; final choices will follow department hosting policy, procurement rules and benchmark results on the target languages.